Packet Tracer - Compare Layer 2 and Layer 3 Devices

Lab Activity & Technical Reference Report

Objectives

- **Part 1:** Compare Layer 2 and Layer 3 Switches
- **Part 2:** Compare a Layer 3 Switch and a Router

Background & Scenario

In this activity, you will use various Cisco IOS commands to examine three different switching topologies and compare the similarities and differences between Cisco Catalyst 2960 (Layer 2) and Catalyst 3650 (Layer 3) switches. You will also compare the routing table of an ISR 4321 router with that of a Catalyst 3650 multi-layer switch.

Part 1: Compare Layer 2 and Layer 3 Switches

Step 1: Explore the Physical Workspace

Physical Port Comparison

Hardware Property	Cisco Catalyst 2960 (ASw-1)	Cisco Catalyst 3650 (D1)
Total Physical Switchports	26 Ports	28 Ports
Fast Ethernet Ports	24 Ports (`FastEthernet0/1` – `0/24`)	0 Ports
Gigabit Ethernet Ports	2 Ports (`GigabitEthernet0/1` – `0/2`)	24 Ports (`GigabitEthernet1/0/1` – `1/0/24`) + 4 SFP Uplinks

Hardware Property	Cisco Catalyst 2960 (ASw-1)	Cisco Catalyst 3650 (D1)
Modular Design	Fixed configuration	Fixed port density with modular power supply/fan bays & uplink options

Questions & Answers

- **Each individual switch has how many physical switchports?**
  ASw-1 (Cisco 2960) has **26 physical switchports**. D1 (Cisco 3650) has **28 physical switchports** (24 RJ-45 ports plus 4 SFP uplink ports).
- **How many Fast Ethernet and Gigabit Ethernet switchports does each switch have?**
  - **ASw-1 (2960):** 24 Fast Ethernet ports and 2 Gigabit Ethernet uplink ports.
  - **D1 (3650):** 0 Fast Ethernet ports and 24 Gigabit Ethernet ports (plus 4 SFP modular ports).
- **List the transmission speed of the Fast Ethernet and Gigabit Ethernet switchports on each switch.**
  - **Fast Ethernet:** 10/100 Mbps.
  - **Gigabit Ethernet:** 10/100/1000 Mbps (1 Gbps).
- **Are either of the two switches modular in design?**
  Neither switch is a fully modular chassis switch (such as a Catalyst 4500 or 6500). However, the **3650 (D1)** features field-replaceable modular power supplies, fan trays, and customizable uplink module slots, whereas the **2960 (ASw-1)** is a completely fixed hardware design.

Step 2: Examine Command Line Interface (CLI) & Operational Differences

Questions & Answers

- **What is the difference between a Layer 2 switch and a Layer 3 switch?**
  A **Layer 2 switch** forwards traffic strictly based on MAC addresses (Data Link Layer) within a single broadcast domain/VLAN. A **Layer 3 switch** (Multi-Layer Switch) can forward traffic based on both MAC addresses (Layer 2) and IP addresses (Layer 3). It performs inter-VLAN routing, supports routed physical ports, and runs dynamic routing protocols directly in hardware using ASICs.

- **What is the difference between a switch's physical interface and the VLAN interface?**
  A **physical interface** corresponds to an actual physical RJ-45 or fiber hardware port on the device used to connect cables. A **VLAN interface** (Switch Virtual Interface or SVI) is a logical Layer 3 interface created in software that represents an entire VLAN; it is primarily used for remote administration or inter-VLAN routing gateway functions.
- **At which layers do 2960 and 3650 switches operate?**
  - **Cisco 2960:** Operates primarily at **Layer 2** (Data Link Layer).
  - **Cisco 3650:** Operates at both **Layer 2 and Layer 3** (Data Link and Network Layers).
- **Do you notice any differences between the output of `show run` on D1 and ASw-1?**
  Yes. On **D1**, `ip routing` is enabled globally, physical interfaces can be configured as routed interfaces via the `no switchport` command, and multiple IP addresses can be assigned to interfaces. On **ASw-1**, all physical interfaces are strictly Layer 2 switchports (`switchport mode access/trunk`), and IP addressing is restricted to the management SVI (`interface Vlan1`).
- **Why do you think the command `show ip route` does not work on ASw-1 but works on D1?**
  `show ip route` displays the IPv4 routing table. Because **D1** is a Layer 3 switch with IP routing activated, it maintains an active routing table. **ASw-1** is a pure Layer 2 switch with no IP routing capabilities enabled, so it does not maintain an IP routing table and rejects or fails to process routing commands.

Part 2: Compare a Layer 3 Switch and a Router

Step 1: Compare D1 (3650 Switch) and R1 (4321 Router)

Questions & Answers

- **Do you notice any similarities or differences between the physical aspects of D1 and R1?**

    - **Similarities:** Both devices possess console management ports, RJ-45 network interfaces, power switches, status LEDs, and metallic rack-mountable chassis.
    - **Differences: R1 (ISR 4321)** has high-density modular expansion slots (NIM slots), removable flash memory ports, and a low count of built-in Gigabit Ethernet interfaces (2–3 routed ports). **D1 (3650)** features a high-density bank of 24 fixed Ethernet ports, dual redundant power supply slots, and dedicated StackWise connectivity ports.
- **What differences do you see between the configurations (`show run`) of R1 and D1?**
    - On **R1**, physical interfaces (e.g., `GigabitEthernet0/0/0`) are Layer 3 routed interfaces by default and accept IP addresses directly.
    - On **D1**, physical interfaces act as Layer 2 switchports by default and require the `no switchport` command before an IP address can be assigned directly to the physical port. D1 also holds configurations for VLAN databases and multiple SVIs.
- **Which command allows configuration of D1 with an IP address on one of its physical interfaces?**
  The **`no switchport`** interface configuration command converts a Layer 2 switchport into a Layer 3 routed interface, enabling the `ip address <ip_address> <subnet_mask>` command.
- **Do you see any similarities or differences between the routing tables (`show ip route`) of R1 and D1?**
    - **Similarities:** Both routing tables display connected (`C`) and local (`L`) routes, subnet masks, next-hop addresses, and exit interfaces using standard Cisco IOS notation.
    - **Differences:** R1's routing table lists physical interfaces and subinterface routes, whereas D1's routing table includes routes associated with SVIs (`Vlan` interfaces) alongside routed physical ports.

Step 2: Compare R2 and D2 Routing & Network Connectivity

Questions & Answers

- **What is present now in the configuration/routing tables of R2 and D2 that was not present in R1 and D1?**

Dynamic routing protocol routes (e.g., **OSPF** designated by `O` or **EIGRP** designated by `D`) or static routes (`S`) are now active, allowing R2 and D2 to exchange network prefixes across the topology.

- **Which network is in the routing table of D2 that was learned from R2?**
  The remote network prefix connected behind R2 (e.g., `10.1.1.0/24` or `192.168.x.x` depending on the specific topology assignment) is listed in D2's routing table as an OSPF/EIGRP learned route.
- **Which device is used to provide communication between networks?**
  A **Layer 3 device** — either a traditional **Router** or a **Layer 3 Switch (Multi-Layer Switch)**.
- **Why were we able to ping across networks without there being a router?**
  Because the **Layer 3 Switch (D1/D2)** has IP routing enabled (`ip routing`) and performs hardware-based inter-VLAN and IP routing directly across its SVIs and routed interfaces using high-speed ASICs.
- **Bonus Question: We say that routers are Layer 3 devices and conventional switches are Layer 2 devices. However, we can assign an IP address to a management interface (SVI) of a Layer 2 switch. How is this possible if switches are Layer 2 devices?**
  An IP address assigned to an SVI (e.g., `interface Vlan1`) on a Layer 2 switch functions strictly as an **host management endpoint**. It allows the switch to process network management traffic directed *to* or *from* the switch itself (such as SSH, Telnet, SNMP, Syslog, or ICMP pings). The Layer 2 switch does **not** perform IP packet forwarding or routing *between* different interfaces or subnets; it merely acts as a network host for administrative access.

Architectural Comparison Summary

Feature / Capability	Layer 2 Switch (e.g., 2960)	Layer 3 Switch (e.g., 3650)	ISR Router (e.g., 4321)
Primary OSI Layer	Layer 2 (Data Link)	Layer 2 & Layer 3	Layer 3 (Network)
Forwarding Engine	MAC Address Table	MAC Table + FIB/TCAM Table	FIB / Routing Table

Feature / Capability	Layer 2 Switch (e.g., 2960)	Layer 3 Switch (e.g., 3650)	ISR Router (e.g., 4321)
Switchport Types	Access & Trunk Ports	Access, Trunk, & Routed Ports	Routed Interfaces / Subinterfaces
Port Density	High (24–48 Ports)	High (24–48 Ports)	Low (2–4 Ports)
Inter-VLAN Routing	Requires external router ("Router-on-a-Stick")	Hardware-based SVIs (Wire-speed)	Software/Hardware Subinterfaces
WAN Technology Support	None	Limited	Extensive (Serial, Cellular, DSL, Fiber)